Recap: Lecture 18

In Lecture 18, we

- Learnt about the *complex Fourier series*, which is to resurface in Topic 6

$$\sum_{n=-\infty}^{\infty} c_n \, \mathrm{e}^{\mathrm{i}k_n x} \quad \text{where} \quad c_n = \frac{1}{2L} \int_{-L}^{L} f(x) \mathrm{e}^{-\mathrm{i}k_n x} \, \mathrm{d}x$$

- Learnt about *Fourier sine/cosine series* and *odd/even extensions*, e.g.,

$$\sum_{n=1}^{\infty} b_n \sin(k_n x) \quad \text{where} \quad b_n = \frac{2}{L} \int_{0}^{L} f(x) \sin(k_n x) \mathrm{d}x$$

  for an odd function $f(x)$.

Lecture 19

In Lecture 19, we will:

1. Work through an example of calculating an anti-derivative Fourier series
2. Work through an example of solving an IVP using Fourier series
3. Introduce Parseval's identity for calculating the average power of a signal

Exercise 4.9a: Find the Fourier series of the odd periodic extension of

$$f(x) = \begin{cases} x, & 0 \leqslant x \leqslant L/2 \\ L - x, & L/2 < x \leqslant L \end{cases}$$

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We seek a Fourier sine series of form $F(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$, where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin(k_n x)\, \mathrm{d}x = \frac{2}{L} \left(\int_0^{L/2} x \sin \frac{n\pi x}{L}\, \mathrm{d}x + \int_{L/2}^{L} (L - x) \sin \frac{n\pi x}{L}\, \mathrm{d}x \right)$$

Therefore

$$b_n = \frac{4L}{n^2\pi^2} \sin \frac{n\pi}{2}$$

which is zero for even n.

The Fourier series is $F(x) = -\frac{4L}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \sin \frac{(2m-1)\pi x}{L}$

Exercise 4.9b: Find a Fourier series for an anti-derivative of $f(x)$

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Use term-by-term integration of the Fourier series of the odd extension of f.

From **Exercise 4.9**a:

$$F(x) = -\frac{4L}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \sin \frac{(2m-1)\pi x}{L}$$

We use

$$\int_0^x -\sin \frac{(2m-1)\pi\xi}{L} \, \mathrm{d}\xi = \frac{L}{(2m-1)\pi} \left[\cos \frac{(2m-1)\pi x}{L} - 1 \right]$$

The integrated Fourier series thus is

$$\int_0^x f(\xi)\,\mathrm{d}\xi = -\sum_{m=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} + \sum_{m=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} \cos\frac{(2m-1)\pi x}{L}$$

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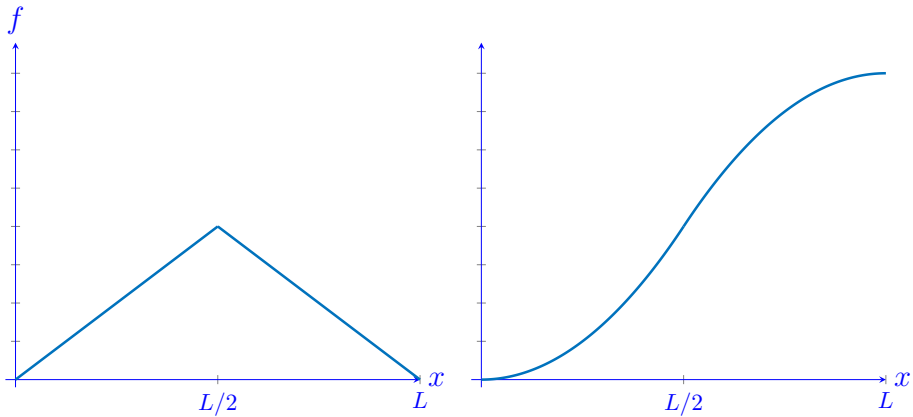
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From direct integration

$$\int_0^x f(\xi)\,\mathrm{d}\xi =$$

RHS shows

$$\begin{cases} x^2/2, & 0 \leqslant x \leqslant L/2 \\ Lx - x^2/2 - L^2/4, & L/2 < x \leqslant L \end{cases}$$



$$\int_0^x f(\xi)\,\mathrm{d}\xi = -\sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} + \sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} \cos\frac{(2m-1)\pi x}{L}$$

How can we calculate the constant term?

$$\int_0^x f(\xi)\,\mathrm{d}\xi = -\sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} + \sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} \cos\frac{(2m-1)\pi x}{L}$$

How can we calculate the constant term?

Integrate the anti-derivative series over $x \in [0, L]$ (a multiple of the cosine period!)

(Integrating a uniformly convergent series gives a uniformly convergent series.)

. . . then the cosine terms go to zero.

Thus

$$-L\sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} = \int_0^L \int_0^x f(\xi)\,\mathrm{d}\xi\;\mathrm{d}x = \frac{L^3}{8}$$

and we have

$$\int_0^x f(\xi)\,\mathrm{d}\xi = \frac{L^2}{8} + \sum_{n=1}^{\infty} \frac{4L^2(-1)^m}{(2m-1)^3\pi^3} \cos\frac{(2m-1)\pi x}{L}$$

Exercise 4.10: Application to an IVP ("A forced oscillator")

Find $y(t)$ such that

$$My''(t) + ky(t) = F(t) \quad \text{where} \quad y(0) = y'(0) = 0$$

where $F(t)$ is the odd periodic extension of (cf. Ex. 4.9)

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Rearrange to standard form

$$y''(t) + \Omega^2 y(t) = \frac{F(t)}{M} = -\frac{4L}{M\,\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \sin \frac{(2m-1)\pi t}{L} \quad \text{where} \quad \Omega^2 = \frac{k}{M}$$

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The homogeneous problem has lin indep solutions $y_1 = \cos(\Omega t)$ and $y_2 = \sin(\Omega t)$

Try a Fourier sine series for the particular solution, i.e.,

$$y_p(t) = \sum_{n=1}^{\infty} b_n \sin(\omega_n t) \quad \text{where} \quad \omega_n = \frac{n\pi}{L} \quad \Rightarrow \quad y_p''(t) = -\sum_{n=1}^{\infty} \omega_n^2 b_n \sin(\omega_n t)$$

assuming term-by-term differentiation. Therefore

$$y_p''(t) + \Omega^2 y_p(t) = \sum_{n=1}^{\infty} b_n(\Omega^2 - \omega_n^2)\sin(\omega_n t) = -\frac{4L}{M\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \sin(\omega_{2m-1} t)$$

Using orthogonality or simply matching coefficients

$$b_{2m-1} = -\frac{4L}{M\pi^2} \frac{(-1)^m}{(\Omega^2 - \omega_{2m-1}^2)(2m-1)^2} \quad \text{and} \quad b_{2m} = 0$$

The general solution is

$$y(t) = C_1 \cos(\Omega t) + C_2 \sin(\Omega t) - \frac{4L}{M\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(\Omega^2 - \omega_{2m-1}^2)(2m-1)^2} \sin(\omega_{2m-1} t).$$

$y(0) = 0$ gives $C_1 = 0$

$y'(0) = 0$ gives (exercise)

$$C_2 = \frac{4L}{M\pi^2\Omega} \sum_{m=1}^{\infty} \frac{(-1)^m \omega_{2m-1}}{(\Omega^2 - \omega_{2m-1}^2)(2m-1)^2}$$

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What happens if there exists an m such that $\Omega = \omega_{2m-1}$?

Let's simplify the ODE to

$$y''(t) + \omega_{2m-1}^2 y(t) = -\frac{4L}{M\pi^2}\frac{(-1)^m}{(2m-1)^2}\sin(\omega_{2m-1}t)$$

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This has solution (exercise)

$$y_p(t) = \frac{2L(-1)^m}{M\pi^2\omega_{2m-1}^2(2m-1)^2}\left\{t\omega_{2m-1}\cos(\omega_{2m-1}t) - \sin(\omega_{2m-1}t)\right\}$$

which grows unbounded in time. This is called resonance!

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In real systems, damping (or some other physical process) will curb this unbounded growth.

Parseval's identity

Given an arbitrary Fourier series $s(x)$, then

$$\big\langle s(x), s(x) \big\rangle = \frac{1}{L} \int_{-L}^{L} \big|s(x)\big|^2 \, \mathrm{d}x = \frac{|a_0|^2}{2} + \sum_{n=1}^{\infty} \left(|a_n|^2 + |b_n|^2\right)$$

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- Can be proved using the orthogonality of the eigenfunctions
- $\sqrt{\langle s(x), s(x) \rangle}$ is the norm/RMS of the signal
- $\langle s(x), s(x) \rangle$ is the average power of a signal

Exercise 4.11: Find average power of periodic extension of x^2 on $(-\pi, \pi]$

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From Ex 4.3, the Fourier series is

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \qquad \text{where} \qquad a_0 = \frac{2\pi^2}{3} \qquad \text{and} \qquad a_n = \frac{4(-1)^n}{n^2}$$

so the average power of the series is

$$\langle x^2, x^2 \rangle = \frac{4\pi^4}{18} + \sum_{n=1}^{\infty} \frac{16}{n^4}$$

Use this result to evaluate the series $\sum_{n=1}^{\infty} \frac{1}{n^4}$

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From direct integration

$$\langle x^2, x^2 \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} x^4 \mathrm{d}x = \frac{2\pi^4}{5}$$

and therefore

$$16 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{2\pi^4}{5} - \frac{2\pi^4}{9} = \frac{8\pi^4}{45} \quad \Rightarrow \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$